

Marcos Esparza

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Summary

Mechanical Engineering student at the University of Texas Permian Basin with hands-on experience in high-power rocketry, experimental thermo-fluids labs, and mechanical systems. Aerodynamic Design Lead for a 10,000-ft COTS competition rocket, using SOLIDWORKS, MATLAB, and OpenRocket for design, analysis, and integration. Seeking a Summer 2026 engineering internship in aerospace or energy systems where I can contribute to testing, analysis, and hardware development.

Education

Bachelor of Science: Mechanical Engineering

Aug 2022 - December 2026 (expected)

University of Texas Permian Basin | GPA: 3.0

Odessa, TX

Relevant Coursework: Thermodynamics II, Fluid Mechanics II, Mechanics of Materials, Thermo-Fluids Lab, Engineering Design

Activities: Falcon Aeronautics & Space Team (FAST), ASME, AIAA, SPE

Technical Skills

Software: SOLIDWORKS, MATLAB, OpenRocket, LabVIEW, Excel

Engineering: Aerodynamic & structural design, test data analysis, systems integration, mechanical assembly, technical writing

Certifications: Tripoli/NAR High-Power Rocketry Level 1; Level 2 in progress

Technical Projects

Falcon Aeronautics & Space Team – Aerodynamic Design Lead & Assembly

- Lead fin-can and airframe design for a COTS competition rocket, targeting ~10k ft apogee with a stable, recoverable flight profile.
- Use SOLIDWORKS, MATLAB, and OpenRocket to size fins, maintain ~1.5–2.0 cal static stability, perform flutter analyses, and integrate recovery and avionics.
- Coordinate with structures, avionics, and recovery subteams to keep the design manufacturable and compliant with competition rules.

Independent High-Power Rocket Certifications (L1 completed, L2 in progress)

- Designed, built, and flew a high-power rocket that achieved ~2,148 ft apogee within ~3% of predicted altitude.
- Currently designing a dual-deploy Level 2 rocket, including airframe layout, recovery system sizing, and avionics bay integration.

Cross-Flow Heat Exchanger Lab – TecQuipment TE93

- Measured airflow velocity and pressure drop across a rod bank using pitot-static and differential pressure transducers.
- Reached ~27 m/s mean airflow at full valve opening and confirmed a linear relationship between dynamic pressure (ΔP_1) and pressure loss (ΔP_2).

Fluid Friction Experiment (H408)

- Ran bend-loss tests on the TecQuipment H408 apparatus for multiple bend radii, measuring head loss and calculating flow, Re, and loss coefficients k and k_L vs Moody/Blasius.

Experience

Logistics Associate — Academy Sports + Outdoors, Midland, TX

Jul 2024 – Present

- Process and stage online and in-store pickup orders, consistently meeting same-day deadlines.
- Perform cycle counts and resolve stock discrepancies to maintain accurate inventory.

Overnight Deli Production — H-E-B, Midland, TX

Oct 2022 – Jan 2024

- Prepared and stocked deli products overnight to support next-day store operations.
- Helped train new team members on shift procedures and safe handling practices.

Team Member — Whataburger, Midland, TX

Oct 2020 – Mar 2022

- Supported front counter and kitchen operations during high-volume periods while maintaining order accuracy and cleanliness.